



December 2025

(For IS)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

11. The sources identify which two primary types of data mining algorithms specifically discussed for use in Social Media?
 A) Regression and Association Rule Mining B) Classification and Clustering
 C) Feature Selection and Dimension Reduction D) Deep Learning and Reinforcement Learning
12. Macro conversions usually refer to:
 A) Major goals like completing a purchase or signing up B) Small actions like clicking a link or viewing a page
 C) Background system performance measures D) Search engine ranking improvements
13. Which statement is true?
 A) Bounce rate = exit rate for all pages B) Exit rate measures overall website traffic
 C) Bounce rate measures single-page sessions, exit rate measures page-specific exit D) Bounce rate is always lower than exit rate
14. Tools such as Tableau Public and KNIME are mainly used for:
 A) Creating advertisements B) Data visualization and advanced analytics
 C) Social media posting automation D) Increasing website loading speed
15. In web analytics, PULSE metrics primarily focus on:
 A) Customer happiness and engagement B) System performance and reliability
 C) Task completion success D) Content quality rating
16. In the context of social media data mining, graph databases are particularly useful because:
 A) They store images and videos more efficiently than relational databases B) They automatically perform clustering without algorithms
 C) They allow modeling of relationships such as friendships and follower networks D) They are designed only for unstructured data like blogs and tweets
17. Which of the following is an example of classification in social media mining?
 A) Predicting whether a tweet is positive, negative, or neutral B) Grouping users into communities based on interaction pattern
 C) Discovering hidden patterns in browsing behavior D) Searching posts containing a specific keyword
18. In clustering algorithms applied to social media:
 A) Each data point is assigned a predefined label B) Clustering works only on numerical datasets, not text or graphs
 C) The algorithm always requires supervised training data D) Data is grouped into clusters based on similarity without prior labels
19. A company wants to automatically detect spam accounts on Twitter. They collect features such as:
 A) Number of followers B) Percentage of users leaving a specific page regardless of how many pages they visited
 C) Ratio of retweets to original tweets D) Presence of suspicious URLs
20. What does exit rate indicate?
 A) Percentage of users leaving after one page B) The total number of clicks on a website
 C) Number of clicks on a page D) Number of returning visitors

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain how performance, audience, sentiment, competitor, paid social, and influencer metrics help businesses improve their digital strategies.	8	L*2	1	2
b)	Develop R programs using ggplot2 to create a box plot and a scatter plot. Illustrate your answer with suitable examples for each type of plot.	8	L3	2	1

IS3334-1

SEE - December 2025

2. a) You are a social media analyst reviewing campaign performance across five platforms: Facebook, Instagram, Twitter, YouTube, and LinkedIn. The following table provides post-level data, including impressions, clicks, likes, shares, comments, reach, and ad spend (for paid posts).
- Calculate the CTR (%) for each platform.
 - Compute the Engagement Score for each platform.
 - Calculate the Engagement Rate (%) for each platform.
 - For paid posts, calculate the Cost per Click (CPC in ₹).
 - Rank the platforms based on:
 - CTR (%) – from highest to lowest.
 - Engagement Rate (%) – from highest to lowest.

Platform	Post Type	Paid (Yes/No)	Impressions	Clicks	Likes	Shares	Comments	Reach
Facebook	Image	Yes(831.60)	4500	230	180	25	12	4200
Instagram	Reel	No	3800	190	210	40	30	3600
Twitter	Text	No	2100	85	95	5	6	2000
YouTube	Video	Yes(1124.7)	9000	500	400	60	35	8500
LinkedIn	Article	No	1200	40	35	3	4	1100

- b) Explain the concept of a DataFrame in R. Develop an R program to import data from a CSV file into a DataFrame and illustrate a few operations on the DataFrame to demonstrate its usage.
3. a) List and briefly explain any three popular analytics platforms used today. And also, distinguish between open-source and licensed analytics platforms with suitable examples, analyzing their advantages and limitations for businesses
- b) Explain how to create a word cloud in R. Write a short example showing how text data can be visualized as a word cloud, and mention its usefulness in data analysis.
- Unit – II**
4. a) How do metrics such as Bounce Rate, Exit Rate, Conversion Rate, Engagement, Macro & Micro Conversions help in on-site and off-site analytics? Explain with example
- b) Explain the concept of Twitter mining. Develop a Python program to fetch recent tweets from a user using the Twitter API with a Bearer token.
5. a) Discuss PULSE and HEART metrics used in social analytics and digital products. Explain the purpose of these metrics in evaluating business and user performance
- b) Explain the concept Twitter Interest Graph. Apply this concept by developing a Python program to fetch recent tweets from a user and convert them into a JSON format containing the tweet ID and tweet text.
6. a) Write a note on ready-made tools for web and social media analytics. Explain the key features of Google Analytics, dashboards, social reports, Tableau Public, and KNIME.
- b) Explain Facebook Graph API and its role in providing structured data for social web mining. Also, describe the purpose of Facebook Apps in accessing user data through the platform.
- Unit – III**
7. a) Explain the concept of keyword search in databases.

IS3334-1

SEE - December 2025

- | | | | | | |
|-------|--|---|----|---|---|
| b) | Explain the motivations, methods, and process of Data Mining in Social Media with examples from social networks and the blogosphere. | 8 | L2 | 5 | 1 |
| 8. a) | Demonstrate text mining on social media data using keyword search and classification techniques. Illustrate with a simple example. | 8 | L3 | 5 | 2 |
| b) | Discuss the importance of query semantics and answer ranking in retrieving relevant results. Illustrate with examples. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
